

RESEARCH ARTICLE

Broadband compact 180° phase shifter based on dumbbell shaped slow wave line

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Abstract

This paper presents a compact 180° phase shifter with flat phase difference in a wide bandwidth. It is a hybrid structure based on a stub-loaded line and a switched-line phase shifter, in which a kind of periodic dumbbell-shaped slow wave line and T-shaped short-ended stub are introduced for miniaturization. The proposed configuration shows a better broadband performance and compact geometric dimension compared with traditional phase shifters. The fractional bandwidths of the proposed structure defined by the return loss of 10 dB and the phase deviation of $\pm 5^\circ$ were 96% and 77%, respectively.

KEYWORDS

broadband 180° phase shifter, dumbbell-shaped slow wave line, miniaturization, quarter-wavelength stub, stub-loaded transmission line

1 | INTRODUCTION

Digital phase shifters play an important role in many applications, such as beam-forming networks, phase-array antennas, radars, and phase modulation communication systems. Broadband constant phase difference and compact size characteristics are usually required in practical applications. It

can be achieved by using various configurations and design methods, of which the most famous one is the Schiffman differential phase shifter.¹ Some improved Schiffman phase shifters are successfully developed by using a 2-layer structure² and a coupled line section with tight coupling.³ The requirements of tight coupling to achieve a wide bandwidth limits their applications more or less. Multilayer structures and narrow coupling gaps slightly increase the cost and difficulty in fabrication and installation.

There is another way to realize broadband phase shifter, which consists of a transmission line (TL) section loaded with open/short stubs. A classical broadband phase shifters without any coupling structure is composed of a fixed $\lambda/2$ TL and $\lambda/8$ parallel open and short stubs.⁴ The phase shifter could achieve $180^\circ \pm 5^\circ$ phase shift with a fractional bandwidth (FBW) of 50%. $\lambda/4$ short-ended stub loaded TLs,⁵ T-shaped open stub loaded TLs,⁶ and multimode resonators⁷ were successfully applied to phase shifter designs to achieve a wide phase range, respectively.

In this letter, a broadband 180° phase shifter with compact size is designed, fabricated, and measured. The proposed structure consists of a reference line and a TL with 3 short-ended stubs, in which a kind of periodic dumbbell-shaped slow wave line (SWL) is introduced for miniaturization. In the following section, performances of the proposed phase shifter and the SWL are discussed and analyzed.

2 | DESIGN OF PHASE SHIFTER BIT

The simplest switched-line differential phase shifter consists of 2 TL branches with different lengths. This structure only provides the required phase difference over a narrow bandwidth. To improve the bandwidth of the phase shifter, short-ended stubs can be added on the shorter TL branch to make the phase slope of 2 TL branches are equal, which means the shorter TL branch will become a band-pass filter (BPF). The bandwidth of the BPF becomes wider with the transmission poles increasing.⁵ A trade-off between the bandwidth of return loss and phase deviation of the phase shifter need to be considered during the designing process. Figure 1 shows the proposed topology for the 180° phase shifter bit consists of a TL branch with characteristic impedance Z_0 , a BPF branch, and 2 SPDT switches. The BPF branch consists of 3 short-ended shunt stubs (TL₁, TL₂, and TL₃) with characteristic impedance Z_{s1} and Z_{s2} , and 2 connecting lines (TL₄ and TL₅) with characteristic impedance Z_m . Electrical

lengths of the short-ended stubs, connecting lines and TL branch are θ_{s1} , θ_{s2} , θ_m , and θ_t , respectively.

First, let $\theta_{s1} = 90^\circ$, $\theta_{s2} = 90^\circ$, $Z_m = 50 \Omega$, $\theta_t = \theta_m \times 2 + 180^\circ$. By tuning θ_m , Z_{s1} , and Z_{s2} , we can get the desired transmission and phase deviation bandwidth.⁸ The overall ABCD matrix of the BPF is a product of 5 individual matrices, that is, 3 short-ended shunt stubs and 2 connecting lines. Subsequently, the S -parameters of the BPF are calculated from the ABCD matrix. Differential phase shift $\Delta\Phi$ for this type of phase shifter is the transmission phase difference between the BPF branch and TL branch. The complex formulations are not shown here for the sake of brevity.

A BPF branch with 5 transmission poles was designed to satisfy $|S_{11}| < -10$ dB, $175^\circ < \Delta\Phi < 185^\circ$, and $\text{FBW} > 70\%$. The overall fractional FBW of the phase shifter is the intersection of the return-loss bandwidth and phase-error bandwidth. The parameters of the circuit are determined as $\theta_m = 180^\circ$, $Z_{s1} = 40 \Omega$, and $Z_{s2} = 33 \Omega$. Figure 2 shows the theoretical calculation results of the reflection of the BPF with 5 transmission poles and the phase difference of the phase shifter. On the basis of this structure, periodic dumbbell-shaped SWLs and bending short-ended stubs are applied to miniaturizing the phase shifter.

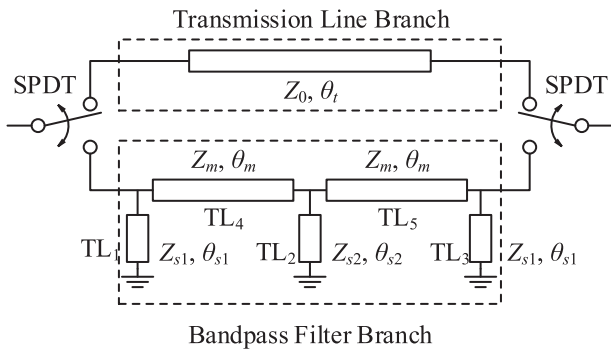


FIGURE 1 Proposed topology for a phase shift bit

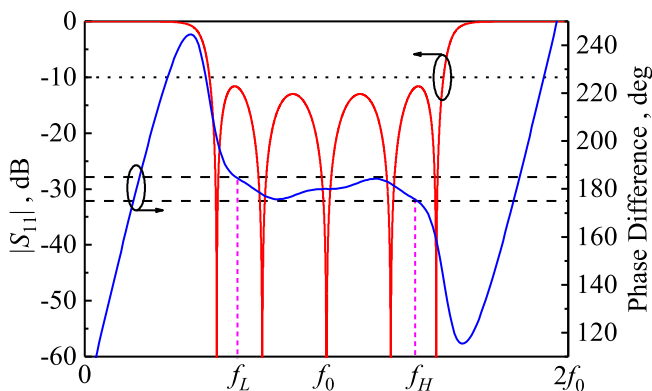


FIGURE 2 Theoretical calculation results of the reflection of the bandpass filter with 5 transmission poles and the phase difference of the phase shifter [Color figure can be viewed at wileyonlinelibrary.com]

3 | DESIGN OF AND MINIATURIZATION

For a given TL, the characteristic impedance Z_c and phase velocity v_p are given by

$$Z_c = \sqrt{L/C} \quad (1)$$

$$v_p = 1/\sqrt{L \cdot C} \quad (2)$$

When L and C rise proportionally, the characteristic impedance keeps the same, while the phase velocity decreases, which means shorter physical length is needed to realize the same electrical length. It is note that both inductance and capacitance of microstrip line are related to line width: inductance increases with decreasing line width and capacitance increases with increasing line width. Thus, a kind of periodic dumbbell-shaped slow wave TL was designed to reduce the physical length of a TL while preserving the electrical properties of the line exactly the same as its conventional structure over a wide bandwidth. The connecting lines in the BPF branch could be taken the place of SWLs to miniaturize the circuit.

The circuit layout of the proposed SWL, consisting of 5 periodic units, is shown in Figure 3. Each periodical unit consists of 2 high-impedance lines, 2 microstrip line parallel-plate capacitors and connecting lines in between. The equivalent lumped circuit model of a periodical unit is shown in the dashed rectangle in Figure 3. L_1 and C_1 represent the inductance and parasitic capacitance of the high-impedance lines, respectively. L_2 and L_3 are induced by the discontinuities and the short connecting lines. C_2 is implemented with 2 microstrip parallel-plated capacitors. The equivalent lumped circuit of the discontinuities between the SWL and 50Ω TL is shown in the dashed circle in Figure 3.

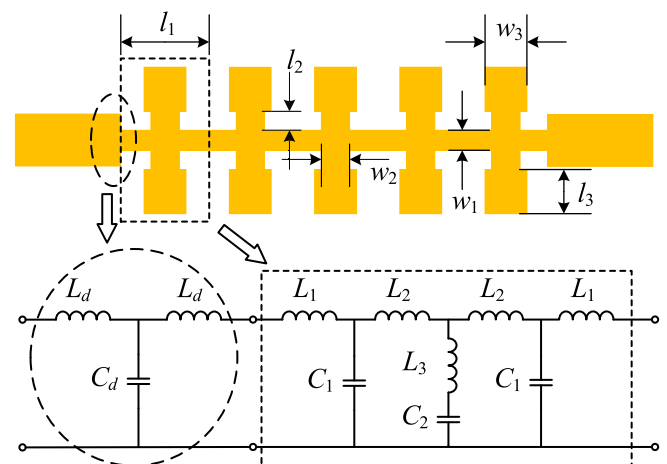


FIGURE 3 Equivalent circuit of the proposed slow wave line [Color figure can be viewed at wileyonlinelibrary.com]

Based on the above analysis of SWL, the design procedure can be described as follows.

Step 1: The total series inductance values and shunt capacitance values can be calculated to achieve the required characteristic impedance and phase shift. For a SWL unit, the capacitance value and inductance value are mainly influenced by L_1 and C_2 .

Step 2: To extract the equivalent inductance and capacitance values, each element is simulated by the EM full-wave simulation and converted to their corresponding the equivalent circuits.

Step 3: The capacitance values can be optimized by tuning l_3 and w_3 , while the inductance values can be adjusted by tuning l_1 and w_1 . However, if w_1 is too small, the insertion loss and the phase slope in the high frequency band will be large. The high frequency characteristics of the SWL can be improved by choosing a large w_1 and adjusting l_2 and w_2 . And then, the dimensions of the SWL still need fine-tuning when considering parasitic couplings between those elements.

Finally, it was decided to use 5 units to achieve 160° phase shift at 2.5 GHz. The dimensions of the SWL are given by $l_1 = 4.3$ mm, $l_2 = 1$ mm, $l_3 = 2.2$ mm, $w_1 = 1$ mm, $w_2 = 1.4$ mm, and $w_3 = 2.1$ mm. The element values of the equivalent lumped circuit model are extracted and summarized in Table 1 for easy reference.

The simulated S -parameters and phase delay of the SWL are shown in Figure 4. The phase delay calculated from the

TABLE 1 Element values of the equivalent lumped circuit model for a $50\ \Omega$ slow wave line

Element	Value (nH)	Element	Value (pF)
L_1	0.76	C_1	0.10
L_2	0.14	C_2	0.40
L_3	0.05	C_d	0.05
L_d	0.11		

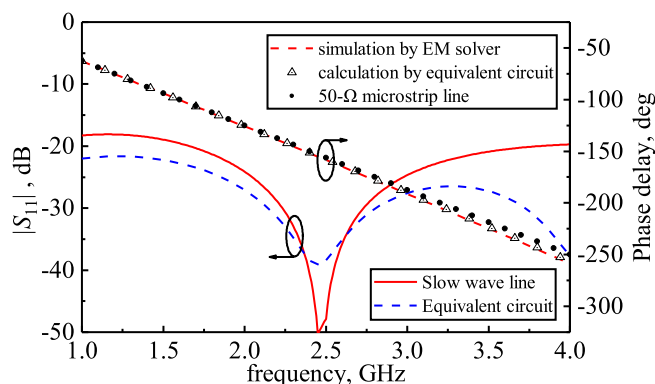


FIGURE 4 Comparison of equivalent circuit and slow wave line [Color figure can be viewed at wileyonlinelibrary.com]

equivalent circuit is consistent with simulation results of the proposed SWL. The electric length of the proposed configuration is 160° at 2.5 GHz, and its physical length is 21.5 mm, which is about two thirds of the conventional $50\ \Omega$ TL with the same electric length. Their phase delay performances are almost identical. Thus, the proposed structure is applied to replacing conventional $50\ \Omega$ lines in the BPF branch for miniaturization.

Among 3 short-ended stubs, impedance of the middle one is low necessitating very broad microstrip TL, which is not suitable to be bent into right angle. Thus, it is transformed into a T-shaped stub as shown in Figure 5, which have smaller size and same performance, when $Z_1 = Z_2 = Z_3/2$, $\theta_3 = \theta_1 - \theta_2$.

4 | EXPERIMENTAL RESULTS AND DISCUSSION

The proposed compact broadband 180° phase shifter is fabricated and measured. A substrate F4B with $h = 1$ mm and a dielectric constant of 2.65 is used. Figure 6 shows the photograph of the fabricated circuit.

The simulated and measured frequency responses of the phase shifter are shown in Figure 7. A good agreement between measurement and simulation is achieved. It has

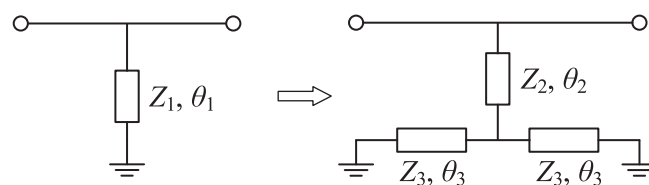


FIGURE 5 T-shaped short-ended stub

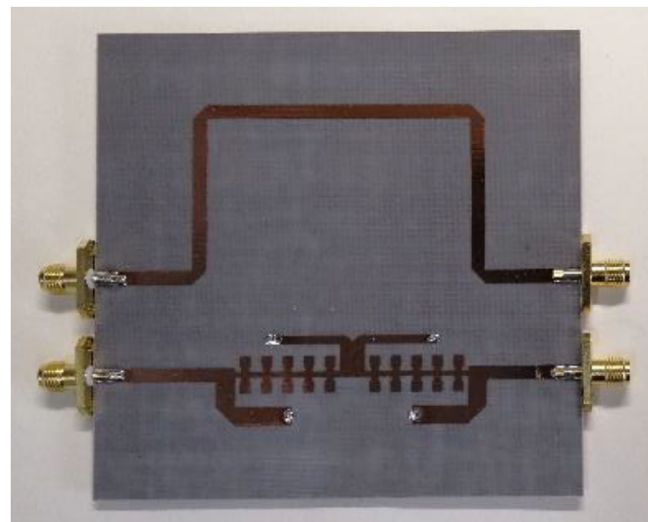


FIGURE 6 Photograph of the fabricated phase shifter [Color figure can be viewed at wileyonlinelibrary.com]

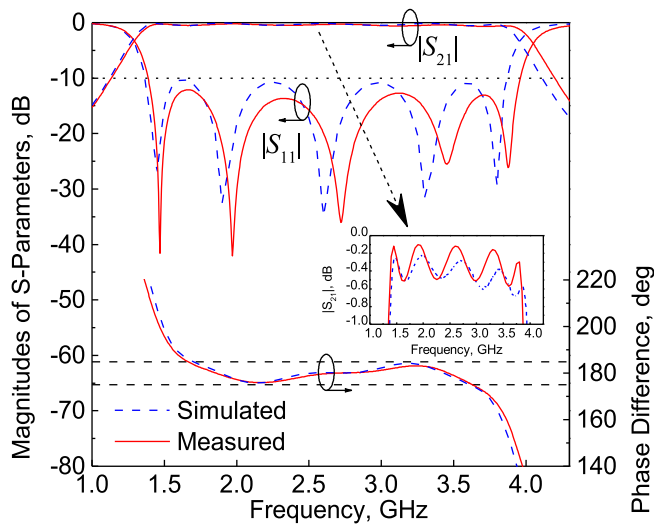


FIGURE 7 Simulated and measured results of 180° phase shifter [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/mop.32403)]

TABLE 2 Comparison of various 180° phase shifters

Reference	RL (dB)	IL (dB)	Phase deviation (deg)	BW (%)	Size ($\lambda_g \times \lambda_g$)
7 I	>12.1	<0.7	± 2	51	0.75×0.07
7 II	>9.2	<1.2	± 3.4	44	0.95×0.12
9	>10	<0.6	± 4.2	40	0.56×0.46
10	>14	<4	± 5	67	1.00×0.25
11	>10	<0.9	± 5	44	0.50×0.38
This work	>10	<0.7	± 5	77	0.70×0.23

been demonstrated that return loss is better than 10 dB and insertion loss is below 0.7 dB from 1.4 to 3.9 GHz. The FBW is 77% with phase deviation within $\pm 5^\circ$. The comparisons with the design in other literatures are shown in Table 2. Compared with other designs, the proposed phase shifter achieves flat phase shift with miniaturized size based on a single-layer structure.

5 | CONCLUSION

In this letter, a compact 180° phase shifter with flat phase difference in a wide bandwidth is proposed. By utilizing the SWL, T-shaped stub and bended short-ended stubs, the dimension of the circuit is reduced while S-parameters and

phase difference remains almost the same. The measured fractional bandwidth of 10 dB return loss and $180^\circ \pm 5^\circ$ phase difference reach 96% and 77%, respectively.

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